# Ethical Dilemmas in Data Privacy: Global Challenges in Cloud Computing

An Engineering Ethics Perspective

# Introduction

- Cloud computing powers modern IT services.
- Creates global access to data storage and processing.
- Raises serious ethical concerns about data privacy.

# What is Cloud Computing?

- On-demand IT services over the internet.
- Includes storage, computing power, networking.
- Types: Public, Private, Hybrid, Community clouds.

# Data Privacy Overview

- Ensuring control over personal data.
- Ethical concern: How data is collected, stored, and shared.
- Involves consent, confidentiality, and transparency.

# Globalization of Data

- Cloud data crosses borders.
- Different countries have different privacy laws.
- Ethical dilemma: Whose law governs the data?

# Ethical Dilemmas in Data Privacy

- Who owns the data in the cloud?
- Should providers be allowed to access user data?
- Is monetizing user data ethically acceptable?

# Cloud Security Risks

- Data breaches, hacking incidents.
- Insider threats or misconfigured security.
- Engineers must address risks ethically.

# Cross-Border Data Flow

- Data stored in various jurisdictions.
- GDPR (EU), CCPA (California), other regional laws.
- Ethical conflict: Legal vs. ethical responsibilities.

# Informed Consent Issues

- Users often don't read or understand data policies.
- Complex Terms of Service lead to uninformed consent.
- Ethics demand clear, simple communication.

# Data Ownership and Control

- Do users or cloud providers control the data?
- Issues with data deletion and management.
- Vendor lock-in raises ethical questions.

# Surveillance and Government Access

- Governments demand access to cloud data.
- National security vs. individual privacy.
- Ethical dilemma: Transparency vs. secrecy.

# Ethical Responsibilities of Engineers

- Implement privacy by design principles.
- Develop secure systems from the ground up.
- Protect user data proactively.

# Transparency and Accountability

- Cloud providers should disclose data use practices.
- Regular audits and compliance checks are needed.
- Accountability for data misuse must be enforced.

# Case Study - Capital One Cloud Breach

- 2019 breach exposed millions of records.
- Resulted from a cloud misconfiguration.
- Ethical lesson: Security is a shared responsibility.

# Case Study - Cambridge Analytica

- Facebook data stored in the cloud was misused.
- Unethical data harvesting and manipulation.
- Showcases the need for strict data governance.

# Legal Frameworks

- GDPR: Focus on user rights and consent.
- CCPA: Protects California residents' data.
- Lack of unified global regulations causes confusion.

# Corporate Social Responsibility (CSR)

- Cloud providers should prioritize data ethics.
- Ethical data handling builds public trust.
- CSR includes privacy and security commitments.

# Future Challenges

- Internet of Things (IoT) and AI increase data volume.
- More data means more risk.
- Ethical frameworks must evolve.

# Technological Solutions

- Use of end-to-end encryption.
- Zero-trust security models.
- Blockchain for data integrity and audit trails.

# Balancing Innovation and Ethics

- Cloud computing drives progress.
- Must align with ethical data protection.
- Long-term trust vs. short-term profit.

# Conclusion

- Ethical data privacy is crucial in cloud computing.
- Engineers, companies, and governments share responsibility.
- Build ethical and secure cloud systems for the future.